

Chitooligosaccharides production from shrimp chaff in chitosanase cell surface display system

Yanli Chen^{a,1}, Zhenmin Ling^{a,1}, Tursunay Mamtimin^a, Aman Khan^a, Liang Peng^a, Jinfeng Yang^a, Gohar Ali^b, Tuoyu Zhou^a, Qing Zhang^a, Jing Zhang^a, Xiangkai Li^{a,b,*}

^a Ministry of Education Key Laboratory of Cell Activities and Stress Adaptations, School of Life Science, Lanzhou University, Tianshuinanlu #222, Lanzhou 730000, Gansu, PR China

^b Key Laboratory for Resources Utilization Technology of Unconventional Water of Gansu Province, Gansu Academy of Membrane Science and Technology, Lanzhou 730020, Gansu, PR China

ARTICLE INFO

Keywords:

Shrimp chaff
Co-fermentation
Chitosanase
Cell surface display
Chitosan
COS production

ABSTRACT

Chitin refers to a natural biopolymer, which is economically significant to next-generation biorefineries. In this study, a novel high-yield method with cell surface-display chitosanase (CHI-1) was built to produce chitooligosaccharides (COS) from shrimp chaff through the co-fermentation in the presence of *Bacillus subtilis* and *Acetobacter* sp. Under the optimized co-fermentation conditions (5 g/L yeast extracts, 10 g/L KH₂PO₄, 6% ethanol, 50 g/L glucose), the final deproteinization (DP) and demineralization (DM) efficiency and the chitin yield were achieved as 94, 92 and 18%, respectively. The engineered *E. coli* BL21-pET23b(+)-NICH1 maintained 81% of the initial enzyme activity after 40 days at room temperature. The crude CHI-1 was inactivated after one-day interacting with prepared chitosan. Moreover, *E. coli* BL21-pET23b(+)-NICH1 still maintained excellent hydrolysis ability in 7 days, and the COS yield reached 41%. Accordingly, the proposed method exhibited excellent stability and a high hydrolysis efficiency to produce COS with whole engineered cells.

1. Introduction

Chitin is recognized as a second most abundant natural biopolymer found in crustaceans, insect exoskeletons, fungal cell walls and some green algae (Younes & Rinaudo, 2015). Chitin exhibits a homopolysaccharide structure formed by the polymerization of *N*-acetyl-D-glucosamine (GlcNAc) through β -(1–4) bonds (Sedaghat et al., 2017). It has three crystal forms (α , β and γ -chitin), among which α -chitin is considered the most abundant and organized by microfibrils arranged in anti-parallel sheets. α -Chitin exists in the cuticle of shrimp and insect exoskeletons, and in the cell walls of fungi. Similarly, β -chitin (parallel sheets) found in squid pens and vestimentiferan worms while γ -chitin (parallel and anti-parallel sheets) was detected in insect cocoons (Younes & Rinaudo, 2015). Chitin is difficult to dissolve in water and other solvents except concentrated acid, which is due to its highly crystalline structure and has limited its applications (Sedaghat et al., 2017).

The shrimp chaff wastes are generated every year after consumption.

The mentioned wastes contain nutritional value components (e.g., vitamins, proteins, and chitin), making shrimp chaff wastes more valuable biological materials (Mathew et al., 2020). Chitin extracted from shrimp chaff is used to produce chitosan, which is further hydrolyzed into water-soluble COS (Xu et al., 2020). It is noteworthy that COS shows many excellent biological activities (such as regulate immune, lower blood fat and blood pressure) while exhibiting the non-toxic effect, biocompatibility and biodegradability (Yuan et al., 2019). COS can also applied in food, sewage treatment, medicine (Wieckiewicz et al., 2017), agriculture (Gabriel Paulraj et al., 2017), chemical and cosmetics (Dimassi et al., 2018) fields.

COS is obtained from chitosan through chemical and enzymatic methods (Ismail et al., 2020). Enzymatic methods include specific enzymes (e.g., chitosanase and lysozyme) and non-specific enzymes (e.g., cellulase, protease and lipase) (Jung & Park, 2014). Chitosanase has two action forms (*endo*-chitosanase and *exo*-chitosanase), mainly derived from actinomycetes, bacteria and fungi (e.g., *Bacillus thuringiensis*, *Bacillus pumilus*, *Streptomyces griseus*, *Bacillus cereus*) (Zhou et al., 2020).

* Corresponding author at: Ministry of Education Key Laboratory of Cell Activities and Stress Adaptations, School of Life Science, Lanzhou University, Tianshuinanlu #222, Lanzhou 730000, Gansu, PR China.

E-mail address: xkli@lzu.edu.cn (X. Li).

¹ These authors contributed equally to this work.

However, chitosanase derived from microorganisms is the crude enzyme solution with low stability and enzyme activity. In addition, the enrichment and purification of chitosanase are complex and expensive. The use of gene cloning and immobilization technology to increase the yield of enzymes and stability. However, there is currently no technology that can combine gene over-expression and immobilization techniques for chitosan hydrolysis (Sinha et al., 2016).

This study proposed a novel method of exploiting the cell surface display technology to express and immobilize specific CHI-1 on surface of *E. coli* for the COS production from shrimp chaff. Compared with crude CHI-1 solution, *E. coli* BL21-pET23b(+)-NICH1 after cultured in a short period could hydrolyze chitosan into COS with higher stability and lower cost (Fig. 1).

2. Materials and methods

2.1. Material

Commercial chitin, chitosan, water-soluble chitosan (deacetylation degree (DD) of 95%) and D-glucosamine were purchased from Sigma-Aldrich in China. The shrimp chaff was provided by Xiaoxu Feed Company in Qingdao, China. The shrimp chaff was cleaned and crushed into powder with a particle size of 0.25 mm to 0.425 mm. *Acetobacter* sp. (BNCC336433) was selected as demineralization bacteria provided by Beina Chuanglian Biotechnology Institute, Beijing, China. Deproteinization bacteria were selected from fermented sludge, including *B. cereus* and *B. subtilis*. These three bacterial strains were preserved in 50% glycerol at -80°C for long time used and -20°C for short time. The

bacterial strains cultured at 30°C , 180 rpm in Luria-Bertani (LB) medium, and prepared 10^8 colony-forming unit (CFU)/mL before use.

2.2. Construction of *E. coli* BL21-pET23b(+)-NICH1 and determination of enzyme activity

B. thuringiensis was screened for CHI-1 and cultured at 37°C , 150 rpm in LB medium for 12 h. The genomic DNA was extracted by using the bacterial DNA extraction Kit (TIANGEN Bio. Inc., China). The primers were designed to amplify the CHI-1 gene based on chitosanase gene of *Bacillus* sp. (Choi et al., 2004). The N-terminal region of the ice nucleation protein (InaK-N) gene from *Pseudomonas syringae* KCTC1832 (NCBI reference sequence no. AF013159) displayed protein or peptide on the cell surface (Liu et al., 2019). The plasmid for cell surface display of CHI-1 gene was constructed by the fusion of InaK-N and CHI-1 genes through overlap PCR. The InaK-NICH1 fragment and pET23b(+) were digested by *Bam*H I and *Xho* I and ligated into the pET23b(+) plasmid, the recombinant plasmid pET23b(+)-NICH1 was transformed into *E. coli* DH5 α for cloning. After confirmation by sequencing, the constructed plasmid pET23b(+)-NICH1 was transformed into *E. coli* BL21 (DE3) for expression. (Liu et al., 2019). *E. coli* BL21 containing pET23b(+) was used as a control.

The constructed *E. coli* BL21-pET23b(+)-NICH1 was cultured and then collected through centrifugation for 2 min at $5867 \times g$ (Sigma Laborzentrifugen, 3K15, Germany). Subsequently, it was washed twice with phosphate buffered saline (PBS) and suspended to 10^9 CFU/mL, 1 mL of the suspension and 1 mL of 1% colloidal chitosan were reacted at 50°C for 30 min. The CHI-1 activity of *E. coli* BL21-pET23b(+)-NICH1

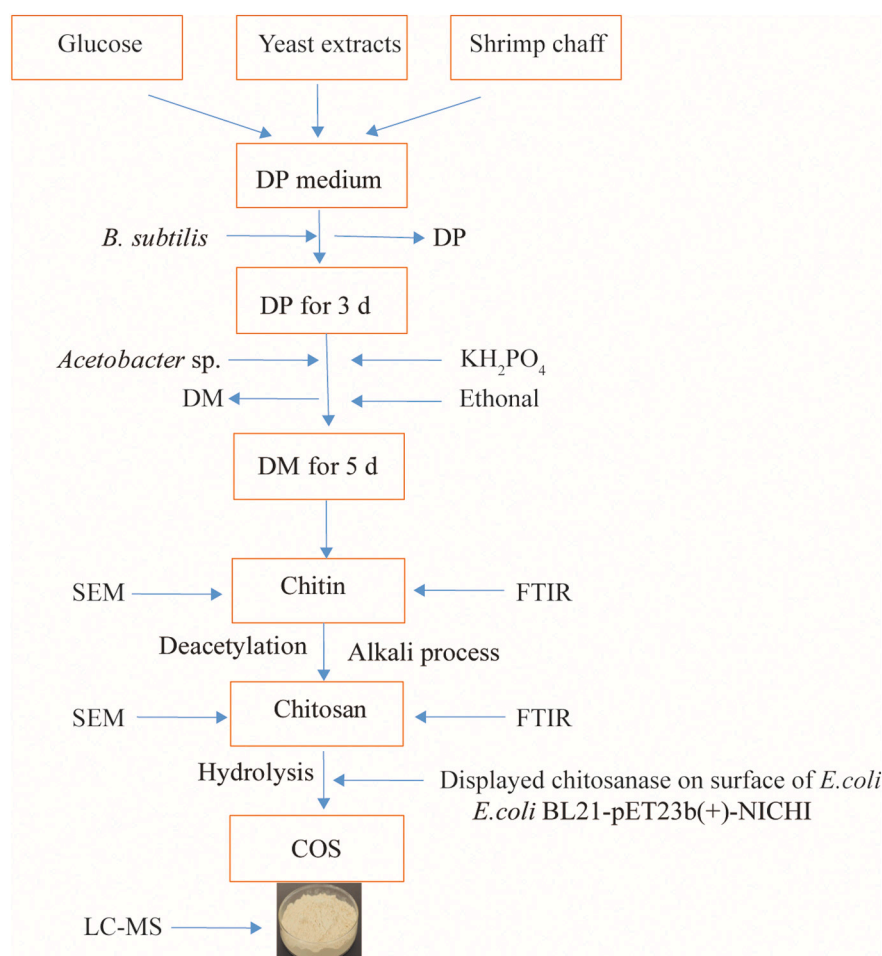


Fig. 1. Schematic of COS extraction by *E. coli* BL21 (+)-NICH1 from shrimp chaff.

was detected according to the dinitrosalicylic acid (DNS) method (Azevedo et al., 2020), one unit of CHI-1 activity of *E. coli* BL21-pET23b(+)–NICH1 activity was defined as the amount of CHI-1 that liberated 1 μmol of reducing sugar per minute. All experiments were repeated three times. 1 mL of suspension of *E. coli* BL21-pET23b(+)–NICH1 recombinants and 1 mL of 1% colloidal chitosan were reacted for 30 min at 25, 30, 35, 40, 45, 50, 55, 60, 65 $^{\circ}\text{C}$, respectively. More basic characteristics of the enzyme-like activation and inactivation energies, inactivation rate constant, half-life, D-, Z-, and Q10 values, enthalpy, entropy, and Gibbs free energy were determined according to the methods (Germec & Turhan, 2019; Germec & Turhan, 2020). The PBS buffer (pH = 3, 4, 5, 6, 7, 8, 9) was used to wash *E. coli* BL21-pET23b(+)–NICH1 recombinants at 10^9 CFU/mL.

2.3. Shrimp chaff composition

Water, ash, chitin and protein contents were detected in shrimp chaff powder. DP and DM efficiency were calculated by the method of the literatures previously reported (Ghorbel-Bellaaj et al., 2012). Shrimp chaff powder (5 g) was dried at 60 $^{\circ}\text{C}$ for 48 h to detect the water content. The 10 g powder was placed in a furnace (MF-1200C, BEQ, China) at 550 $^{\circ}\text{C}$ for 6 h, the remaining residue was the ash content. The total nitrogen content was determined by the Kjeldahl nitrogen analyzer (K9840, Hanon, China). Protein content was calculated by multiplying subtraction of total nitrogen and chitin nitrogen by 6.25. Chitin content was determined by subtracting protein and ash from the dried shrimp chaff powder. DP and DM efficiency were calculated by the following Eqs. (1) and (2), respectively.

$$\text{DP (\%)} = [1 - (M_F \times P_F) / (M_0 \times P_0)] \times 100 \quad (1)$$

$$\text{DM (\%)} = [1 - (M_F \times A_F) / (M_0 \times A_0)] \times 100 \quad (2)$$

where M_0 and M_F were the weight of original samples and fermented residues, g; P_0 and P_F were the percentage of protein in original samples and fermented residues, %; A_0 and A_F were the percentage of ash in original samples and fermented residues, %, respectively.

2.4. Selection of DP bacteria

B. cereus and *B. subtilis* were inoculated into 100 mL fermentation medium (50 g/L shrimp chaff, 50 g/L glucose, 1 g/L yeast extracts) at an inoculum of 3% in a 250-mL flask, and then cultivated at 30 $^{\circ}\text{C}$, 180 rpm, respectively (Zhang et al., 2021). The fermented supernatant (500 μL) was collected at the end of operation, then protease activity and pH were measured. The other fermented liquid was centrifuged at $13,201 \times g$ (Sigma Laborzentrifugen, 3K15, Germany) for 10 min, then the residue was washed and dried at 60 $^{\circ}\text{C}$ to detect the DP%. The protease activity was measured based on the method of Zhang et al. (2021), the tyrosine amount formed in the enzyme reaction was measured based on absorbance at 680 nm. One enzyme activity unit was defined as the amount of enzyme that produced 1 μg of tyrosine per minute under certain conditions. Subsequently, the two bacteria (*B. cereus* and *B. subtilis*) were co-fermented with *Acetobacter* sp. respectively to measure titratable acid (TTA) using 0.1 M NaOH to titrate to pH 8.0. Ca^{2+} was measured by the atomic absorption method (Dun et al., 2019).

2.5. Optimization of co-fermentation conditions

The influence of the medium initial composition (glucose 0, 10, 30, 50, 70, 90 g/L; yeast extracts 0, 3, 5, 7, 9 g/L) on DP and DM efficiency was investigated for co-fermentation of strain *B. subtilis* and *Acetobacter* sp. by the one-factor-at-a-time method. *B. subtilis* was fermented for 3 days to remove proteins of shrimp chaff in the initial medium, then 5 g/L KH_2PO_4 , 6% ethanol and 5% *Acetobacter* sp. were added and co-fermented for another 5 days at 30 $^{\circ}\text{C}$, 180 rpm. In addition, single

fermentation of *B. subtilis* was performed during different fermentation times (2.0, 2.5, 3.0, 3.5 and 4.0 days) after the optimum concentrations of glucose and yeast extracts were determined. The influence of KH_2PO_4 (0, 2.5, 5, 7.5, 10, 12.5 g/L) and ethanol (2, 4, 6, 8, 10 and 12%, v/v) on DP and DM efficiency was also studied by the one-factor-at-a-time method. The crude chitin obtained through the fermentation was repeatedly filtered, washed and dried at 60 $^{\circ}\text{C}$, then treated with hydrogen peroxide (10%, v/v) at the ratio of 1:10 for 2 h at 60 $^{\circ}\text{C}$ to prepare chitosan (Zhang et al., 2021).

2.6. Molecular weight and DD of chitosan and chitin

The fermented chitin was deacetylated (Nwe et al., 2014) to produce chitosan. Potentiometric titration was used to determine DD. Chitin (0.25 g) was dissolved in 30 mL HCl and then stirred for 30 min, followed by titrated with 0.1 mol/L NaOH. The pH values were read when every 1 mL NaOH solution was added until the pH was neutral. Chitin was dissolved to 5% LiCl/N, N-dimethylacetamide (DMAC) to detect molecular weight by using an Ubbelohde viscometer (QI HANG, 01130001, China) at 25 $^{\circ}\text{C}$. The molecular weight was calculated by the Mark–Houwink–Sakurada (MHS) equation (Wang et al., 1991):

$$[\eta] = K (M_v)^{\alpha}$$

where K and α denote the constant values of 7.6×10^{-5} dL/g and 0.95.

Chitosan was dissolved in 0.1 M acetic acid and 0.2 M sodium chloride, and the viscosity of chitosan was determined (Sedaghat et al., 2017).

Where K and α denote the constant values of 1.81×10^{-3} (cm^3/g) and 0.935.

2.7. Analysis of structural features

The structural characteristics of chitin and chitosan samples were analyzed with scanning electron microscope (SEM) and Fourier transform infrared spectroscopy (FTIR) (Teli & Sheikh, 2012).

2.8. Analysis of degradation products and efficiency of chitosan by *E. coli* BL21-pET23b(+)–NICH1

The relative enzyme activity in 40 days was measured by taking 1 mL of *E. coli* BL21-pET23b(+)–NICH1 suspension (10^9 CFU/mL) to react with prepared chitosan. The same goes for *B. thuringiensis* cultured supernatant (crude CHI-1). To compare the effectiveness of hydrolysis between free CHI-1 and *E. coli* BL21-pET23b(+)–NICH1. *E. coli* BL21-pET23b(+)–NICH1 suspension (10^9 CFU/mL) and crude CHI-1 reacted with the prepared chitosan at 50 $^{\circ}\text{C}$ for 1–7 days, respectively. The reaction mixture was centrifuged for 15 min at $13,201 \times g$ (Sigma Laborzentrifugen, 3K15, Germany) and filtered with 0.22 μm membrane (MCE Syringen Filter, Beijing Labgic Technology Co., Ltd., China). 5 μL of filtered reaction mixture was injected into liquid chromatography and mass spectrometry (LC-MS) system by using Hamilton syringe (HAMILTON Bonadzu, Switzerland). The liquid phase consisted of 70% acetonitrile and 30% MilliQ water and the flow rate is 1.0 mL/min. The absorption values with m/z range from 300 to 1100 were monitored at 210 nm (Guo et al., 2019). Next, the production rate of COS was calculated after the hydrolysis for different times to detect the ability to hydrolyze chitosan at different concentrations.

2.9. Statistical analysis

All experiments were performed in triplicate, and the data were expressed as mean $\pm$ standard deviation (SD). The difference was evaluated by analysis of variance (ANOVA) based on Duncan's multiple range test via the software of SPSS version 26.0. The value of P below 0.05 was accepted as significant difference.

3. Results and discussion

3.1. Basic characteristics of CHI-1

Chitosanases (EC 3.2.1.132), belonging to glycoside hydrolases (GHs), could hydrolyze β -(1-4)-glycosidic linkages and the similarity of amino acid sequence, CHI-1 is classified in 7 GH families (Monaghan et al., 1973). *B. thuringiensis* was selected from the previous co-fermentation system with shrimp chaff and corn stalks to enhance the methane system (Ali et al., 2021). CHI-1 secreted by *B. thuringiensis* belongs to GH8 families (Fig. S1).

E. coli BL21 displayed CHI-1 on the surface and effectively degraded chitosan as compare to the control (Figs. 2A, S2). CHI-1 was successfully anchored to the cell surface by ice nucleoprotein and expressed the CHI-1 activity. The CHI-1 activity of *E. coli* BL21-pET23b(+)-NICH1 and crude CHI-1 were measured under different pH values (3, 4, 5, 6, 7, 8, 9) and temperatures (25, 30, 35, 40, 45, 50, 55, 60 and 65 °C) (Fig. 2C, D). The optimum temperature (50 °C) and pH (7) were detected for *E. coli* BL21-pET23b(+)-NICH1 and crude CHI-1. The other basic characteristics were also determined (Tables S2, S3, S4, S5 and S6). This demonstrates that the display technique maintains the basic enzymatic characteristics of CHI-1. The suitable temperature range for other chitosanases is also relatively high, generally between 50 and 60 °C (Ding et al., 2020; Naqvi et al., 2016). The optimum pH of CHI-1 has similar with chitosanase from strain *Acinetobacter* CHB10176 (Monaghan et al., 1973). Other chitosanases have been reported to have different sensitivities for metal ions (Monaghan et al., 1973; Zhu et al., 2019). In this experiment, the effect of 0.5 mM several metal ions (FeSO_4 , CuCl_2 , CaCl_2 , MnSO_4 , MgSO_4 , ZnCl_2 , FeCl_3) (Fig. 2B) on CHI-1 activity of *E. coli* BL21-pET23b(+)-NICH1 and crude CHI-1 was detected. Mn^{2+} , Mg^{2+} and Fe^{3+} showed promotion effect on *E. coli* BL21-pET23b(+)-NICH1 and crude CHI-1 while Zn^{2+} inhibited. Compared to crude CHI-1, *E. coli* BL21-pET23b(+)-NICH1 improved the tolerance of the enzyme to metal

ions. Only chitosan could be hydrolyzed by *E. coli* BL21-pET23b(+)-NICH1 recombinants in several macromolecules (Fig. 2A) connected by glycosidic bonds. The CHI-1 of *E. coli* BL21-pET23b(+)-NICH1 activity of degrading colloidal chitosan was 76.3% of water-soluble chitosan, which demonstrated that large molecules were first hydrolyzed into smaller molecules and then hydrolyzed into COS (Fig. S3).

3.2. Selection of deproteinization bacteria

The basic chemical compositions of shrimp chaff were analyzed. The contents of water, ash, protein and chitin were 20, 12, 45 and 23%, respectively (Table 1). The protease activity (135 U/mL) in *B. subtilis* group was slightly higher than *B. cereus* (127 U/mL) group on the fifth day by exploiting shrimp chaff as a substrate (Fig. 3A). The initial pH value of the fermentation broth was 7.2 and dropped to pH 6.3 on the fifth day in the *B. subtilis* group, the pH of the *B. cereus* group dropped to pH 6.8 (Fig. 3B). The DP efficiency was 87% and 81% for *B. subtilis* and *B. cereus* group after 5 days (Fig. 3C). The TTA of the final fermentation supernatant was measured to increase to 39 g/L in co-fermentation system of *B. subtilis* and *Acetobacter* sp. group (Fig. 3D). Obviously, in co-fermentation system of *B. cereus* and *Acetobacter* sp., TTA was not detected. In the *B. subtilis* and *Acetobacter* sp. co-fermentation, the concentration of Ca^{2+} in the supernatant was 5 g/L after 8 days, which is higher than 1 g/L of the *B. cereus* treatment group (Fig. 3E). The DM efficiency (Fig. 3F) of *B. subtilis* and *Acetobacter* sp. co-fermentation group was 88%, higher than 9% of *B. subtilis* and *Acetobacter* sp. co-fermentation.

3.3. Optimization conditions for co-fermentation

The different concentrations of glucose and yeast extracts were added to the fermentation system to test the effect on the growth of *B. subtilis* and then affected the DP% and DM%. The DP% and DM% were

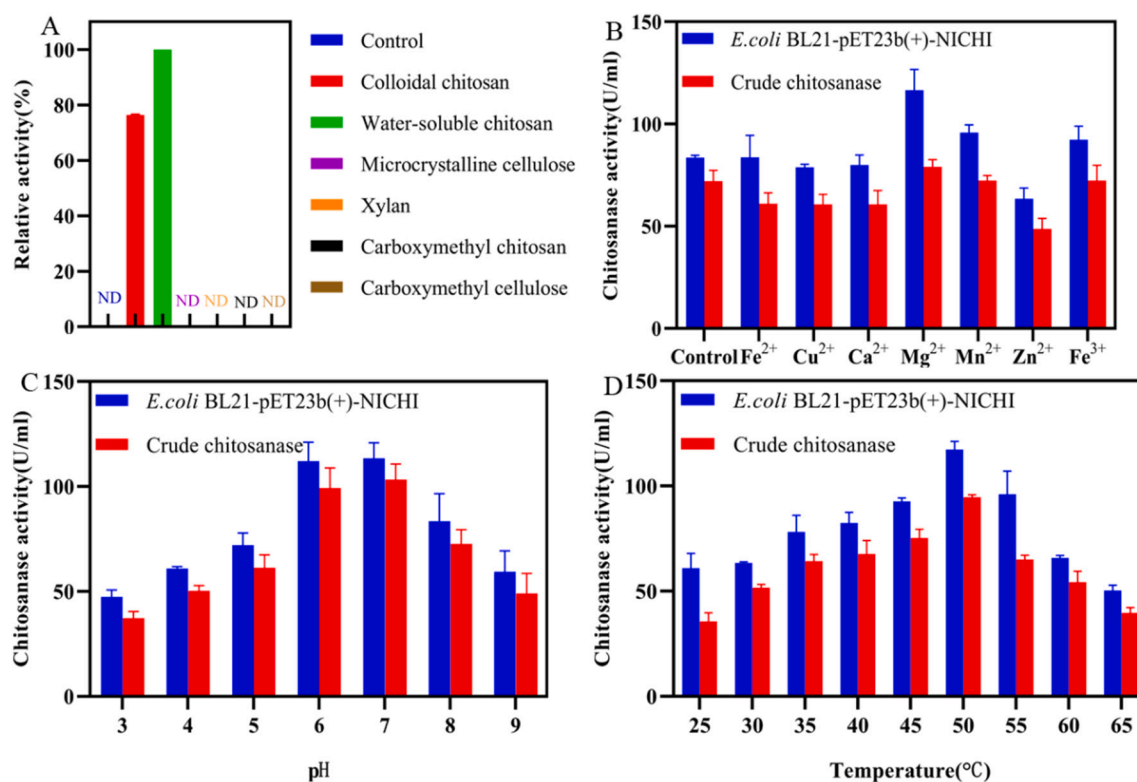


Fig. 2. (A) The relative hydrolysis efficiency of different substrates is hydrolyzed by *E. coli* BL21-pET23b(+)-NICH1. Control presents the hydrolysis of water-soluble chitosan by *E. coli* BL21-pET23b(+). (B) Effect of several metal ions on activity of *E. coli* BL21-pET23b(+)-NICH1 and crude CHI-1. (C, D) Effect of different temperature and pH on activity of *E. coli* BL21-pET23b(+)-NICH1 and crude CHI-1. All values are presented as mean $\pm$ SD based on three parallel experiments ($n = 3$).

Table 1
Physicochemical parameters of shrimp chaff powder, chitin and chitosan.

Samples	Ash (%)	Protein (%)	Water (%)	Chitin/chitosan (%)	Mw (kDa)	DD (%)
Shrimp chaff powder	19.7 ± 0.2	45.3 ± 0.3	12.2 ± 0.1	22.7 ± 0.2	ND	ND
Chitin prepared in this study	4.2 ± 0.2	7.7 ± 0.1	9.0 ± 0.2	79.1 ± 0.7	53.6 ± 0.5	14.3 ± 0.3
Chitosan prepared in this study	ND	ND	0.2 ± 0.01	99.8 ± 0.8	11.1 ± 0.2	95.0 ± 0.4

Mw: molecular weight. DD: deacetylation degree. ND: not determined. All values were expressed as mean ± SD based on three parallel experiments (n = 3).

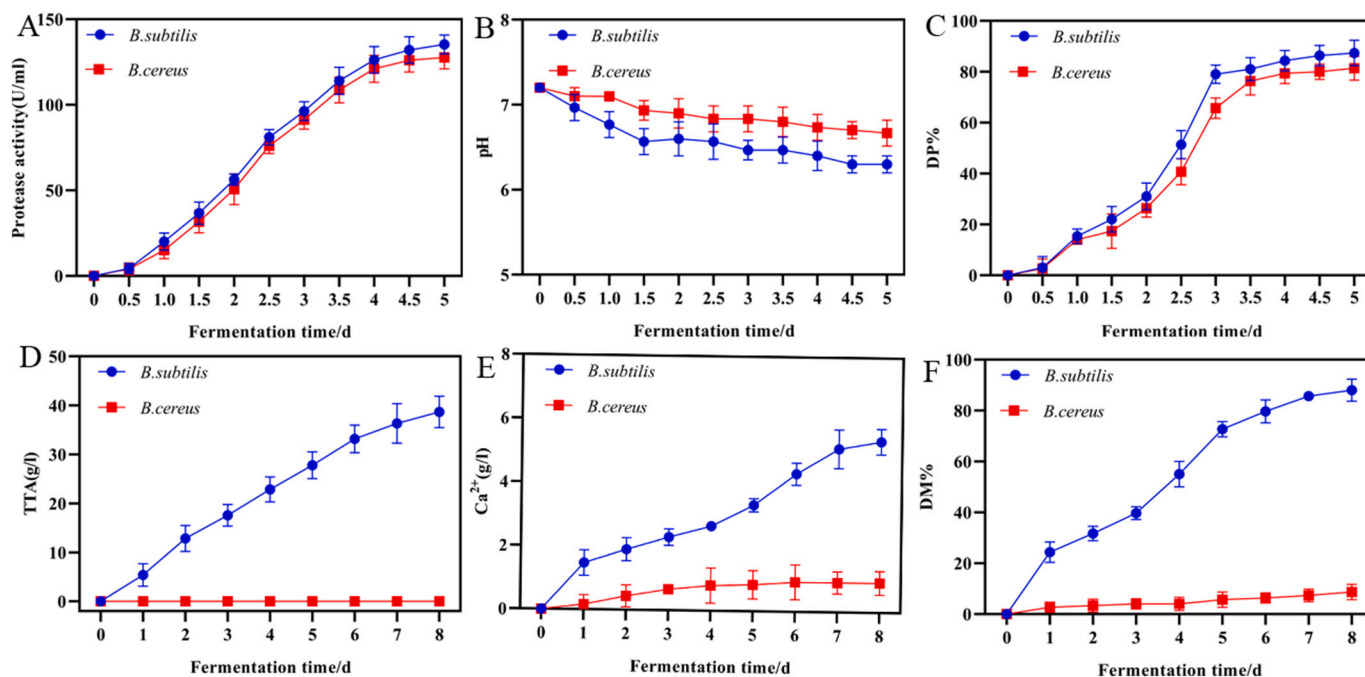


Fig. 3. Changes in protease activity (A), pH (B), DP efficiency (C) of *B. subtilis* and *B. cereus* culture. Changes in acetic acid production (D), Ca^{2+} concentration (E), and DM efficiency (F) in the successive co-fermentation culture. Two protease-producing bacteria (*B. subtilis*, *B. cereus*) were fermented in the initial medium for 3 days, then *Acetobacter* sp. was cultured for another 5 days after feeding 6% (v/v) ethanol. All values are presented as mean ± SD based on three parallel experiments (n = 3).

about 91% and 89% when glucose concentrations were 50, 70 and 90 g/L (Fig. 4A). The DP% and DM% were 91% and 88% at 5 g/L and 7 g/L of yeast extracts (Fig. 4B). In contrast, the DP% and DM% were lower at 9 g/L of yeast extracts. More yeast extracts stimulated the overgrowth of *B. subtilis* to inhibit the growth of *Acetobacter* sp. (Lombard et al., 2014). 0.04–40 g/L KH_2PO_4 was suitable for the microbial growth (Rywinska et al., 2015). It (Fig. 4D) presents the effect on the DP% and DM% with 5 g/L KH_2PO_4 or without 5 g/L KH_2PO_4 . The presence or absence of KH_2PO_4 didn't affect DP%, but obviously affected the DM% situation. Accordingly, different concentrations of KH_2PO_4 (0, 2.5, 5, 7.5, 10 and 12.5 g/L) were added on the 3rd day to detect the effect on DP% and DM% (Fig. 4E). The DP% was the best at 10 g/L KH_2PO_4 . The DM% was same at 10 g/L KH_2PO_4 and 12.5 g/L KH_2PO_4 . Therefore, 10 g/L KH_2PO_4 was the best for both DP% and DM%. Likewise, different concentrations of ethanol (2, 4, 6, 8, 10 and 12%) were added to the fermentation system on the 3rd day (Fig. 4F). 6% and 8% ethanol were the best concentrations for DM%, 2, 4 and 6% ethanol were the best concentrations for DP%. In a word, 6% ethanol was the best concentration for both DP% and DM%.

3.4. SEM and FTIR analysis

The qualitative crystalline microfiber structure could not be observed on the surface of the shrimp chaff at 20,000 times of magnification (Fig. 5A). The surface of single fermentation chitin of *B. subtilis* was smooth and porous without any excessive impurities, and the

crystalline microfiber structure could be observed (Fig. 5B). Compared with Fig. 5B, it (Fig. 5C) showed that the surface of co-fermentation chitin was smoother and had more pores, which suggested that co-fermentation could be better than single fermentation to remove proteins and mineral salts. There was no crystalline microfiber structure on the surface (Fig. 5D). Compared with B and C, the structure of chitosan was not continuous and compact, whereas the porous structure was remained, which could be convenient for microorganisms or enzyme to absorb and could easily be degraded (Choi et al., 2004; Ding et al., 2020).

The three mentioned substances were detected at the wavelength of 400–4000 cm^{-1} . The FTIR spectra of the samples in this study (Fig. 5E) were in line with the previous report results (Affes et al., 2019). For a and b, the absorption peaks of a and b at 3267 cm^{-1} represented the tension vibration (NHCOCH_3) of the NH bond. 1657 cm^{-1} was the absorption peak of stretching vibration of the amide I groups (C=O). 1559 cm^{-1} was amide II, which was the absorption peak of C=N stretching and N–H bending. The band at 1158 and 1075 cm^{-1} was attributed to the —C—O—C stretching vibration modes. At nearly 2873 cm^{-1} , the characteristic signal of the —CH_2 stretching was detected. 1379 cm^{-1} suggested the vibration stretching of —CH_3 , 3424 cm^{-1} in the observed FTIR spectral of a band corresponded to —NH_2 of chitosan (Kavaz et al., 2019).

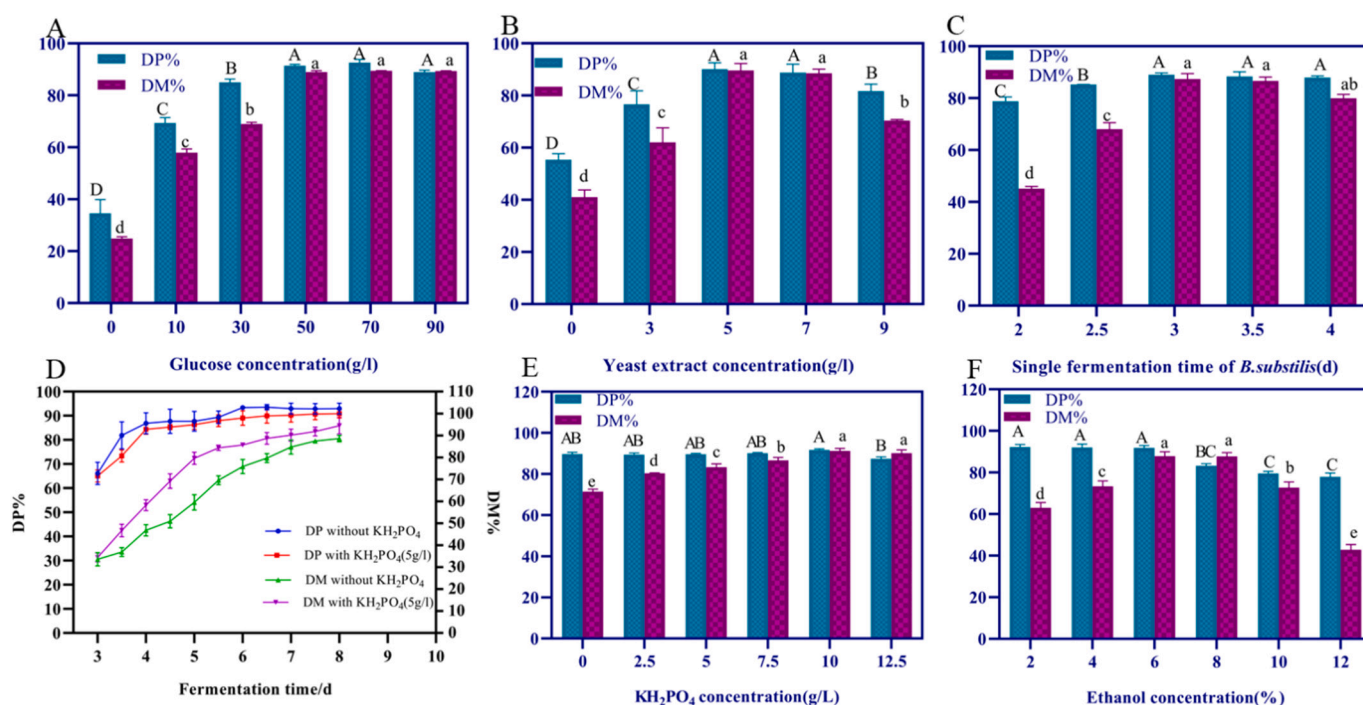


Fig. 4. Effect of glucose concentration (A), yeast extract concentration (B), KH_2PO_4 (D, E) and ethanol concentration (F) on DP and DM efficiency in the successive co-fermentation culture. *B. subtilis* was fermented in the initial medium for 3 days, then *Acetobacter* sp. was cultured for another 5 days. Single fermentation of *B. subtilis* (C) on DP and DM efficiency in 4 days. All values were expressed as mean $\pm$ SD based on three parallel experiments ($n = 3$). Different letters above the bars indicate significant differences ($p < 0.05$) between the groups.

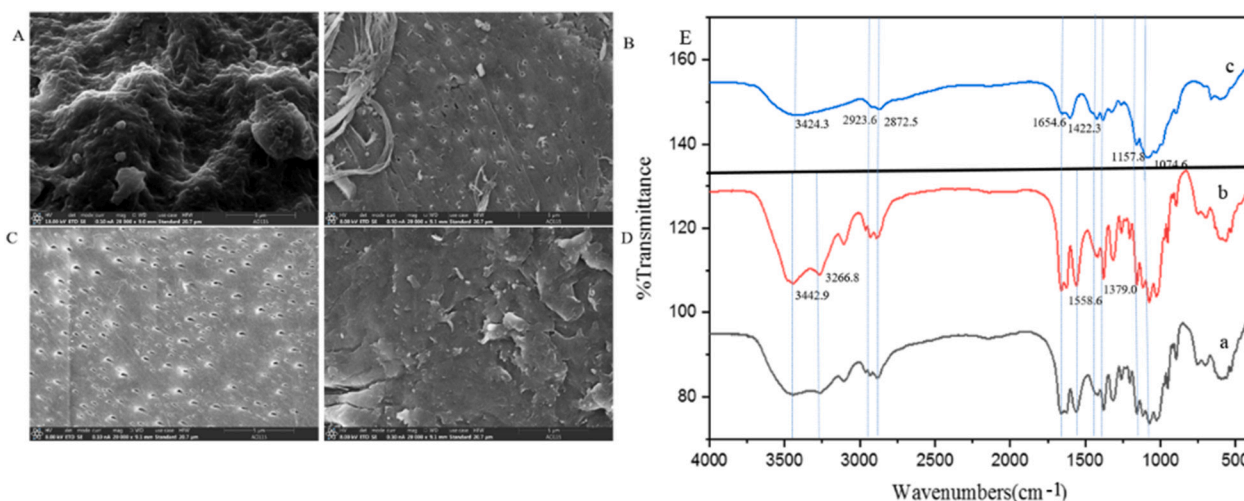


Fig. 5. SEM images of shrimp chaff, chitin and chitosan samples. A: shrimp chaff; B: chitin prepared by the single fermentation of *B. subtilis* for 4 days; C: chitin prepared by the successive co-fermentation with *B. subtilis* and *Acetobacter* sp.; D: chitosan prepared by the deacetylation of chitin prepared by the successive co-fermentation with *B. subtilis* and *Acetobacter* sp. FTIR spectra of sample (E). a: commercial chitin; b: chitin prepared by the co-fermentation of *B. subtilis* and *Acetobacter* sp.; c: chitosan prepared by the deacetylation of chitin prepared by the successive co-fermentation with *B. subtilis* and *Acetobacter* sp.

3.5. Hydrolysis efficiency and products

The culture of *B. thuringiensis* was centrifuged, and the supernatant was acted as the crude CHI-1 solution. As indicated from the results, *E. coli* BL21-pET23b(+)-NICH1 recombinants exhibited a higher CHI-1 stability compared with crude CHI-1. At room temperature, *E. coli* BL21-pET23b(+)-NICH1 maintained nearly 81% of the initial CHI-1 activity after 40 days (Fig. 6A). However, the crude CHI-1 was about 32% of the initial enzyme activity after 40 days. The crude CHI-1 was inactivated after one-day reaction with prepared chitosan and the COS production rate reached the highest value of 9%. In addition,

chitosanase from *B. pumilus* BN-262 achieved a very low COS yield and a short effective hydrolysis period (Kuroiwa et al., 2009). The concentration of *E. coli* BL21-pET23b(+)-NICH1 was at the 10^9 CFU/mL, the hydrolysis ability of chitosan was higher than the crude CHI-1 (Fig. 6B). Moreover, *E. coli* BL21-pET23b(+)-NICH1 exhibited an excellent efficiency among 1–7 days and the COS yield was 41% on the 7th day. It suggested the stability of CHI-1 was increased by cell surface of the *E. coli* BL21 displayed CHI-1 (Fig. 8B). The effect of this technique on the stability of CHI-1 was similar to the chitosanase immobilized on DEAE-cellulose (Zheng & Xiao, 2004). It illustrated that chitosanase was displayed on the cell surface of the *E. coli* BL21, which could be considered

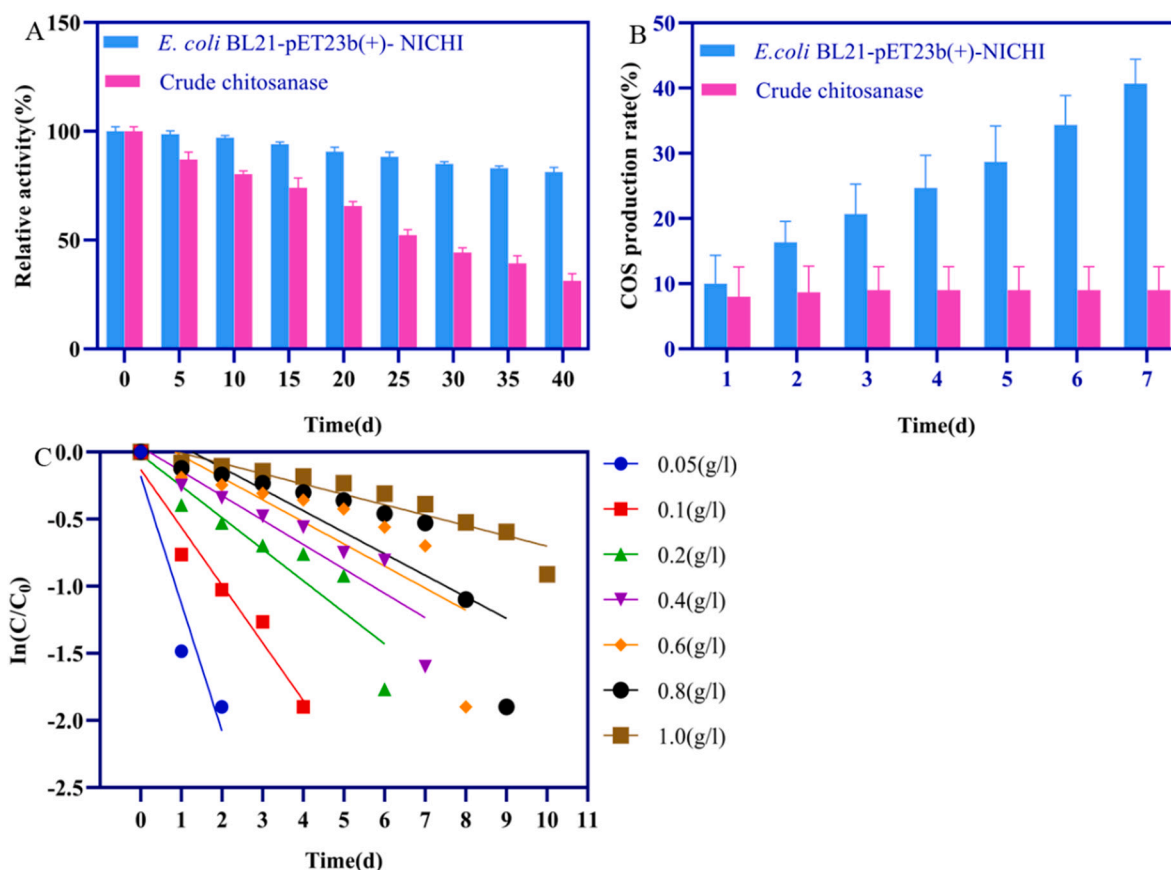


Fig. 6. Relative enzyme activity of crude CHI-1 and *E. coli* BL21-pET23b(+)-NICH1 in 40 days (A). COS yield after the hydrolysis of chitosan (1.0 g/L) with crude CHI-1 and *E. coli* BL21-pET23b(+)-NICH1 in 1–7 days (B); Kinetics of the hydrolysis of chitosan (0.05–1.0 g/L) (C). All values were expressed as mean $\pm$ SD based on three parallel experiments ($n = 3$).

as a whole-cell catalyst to maintain the stability of CHI-1. Compared with the conventional collection and purification of chitosanase (De Araujo Padilha et al., 2016; Nidheesh et al., 2015) and the immobilization of chitosanase, the separation and purification steps of the enzyme were omitted and the cost was reduced. *E. coli* BL21-pET23b(+)-NICH1 could be used in the industry at large-scale application for the COS production. Fig. 6C further illustrated that the lower the concentration, the faster the hydrolysis could be.

Chitosan with DD of 95% and the average molecular weight of 11.1 kDa was hydrolyzed for 3 days and 7 days at 50 °C by *E. coli* BL21-pET23b(+)-NICH1 suspension and crude CHI-1, respectively. The LC-MS analysis for hydrolyzed products was performed at the m/z range of 300–1100 to identify COS (Zhang et al., 2015). GlcN2 (340.9), GlcN3 (520.1), GlcN4 (684.4), GlcN5 (847.4) and GlcN6 (1007.1) were all detected (Figs. 7A, B, C, S4). The content of each oligomer was determined (Table S1). In the previous study (Li et al., 2021), they used a temperature-induced promoter to induce the expression of fusion protein. However, in our study, there was no other promoter was inserted to pET23b(+) to induce the protein expression. Furthermore, our study improved a novel and simple way to prepare COS from shrimp chaff, which could be promising in the biotechnology application for COS preparation from shrimp chaff. CHIS5 (Guan et al., 2020) obtained through the marine metagenome analysis method exhibited the identical degradation mode, and COS could be extensively used in industry, medicine and other fields (Li et al., 2020; Muanprasat & Chatsudthipong, 2017; Yang et al., 2016). COS could facilitate plant growth and nutrient absorption (Chatelain et al., 2014), thereby alleviating the salt stress of wheat seedlings (Zhang et al., 2018). GlcN6 showed the potential to inhibit angiogenesis (Xiong et al., 2009). COS could protect human embryonic liver cells from oxidative stresses induced by

hydrogen peroxide, thereby inhibiting the oxidation of DNA and proteins and inhibiting cell apoptosis by regulating related proteins (Park et al., 2004; Xiong et al., 2009).

4. Conclusions

A GH8 CHI-1 from *B. thuringiensis* was successfully expressed on the cell surface of *E. coli* BL21 maintained basic enzymatic characteristics. It effectively hydrolyzed the chitosan with DD of 95% to produce COS with a value-added effect. The crude CHI-1 was almost inactivated after one-day reaction with chitosan at the optimum temperature and pH. However, the CHI-1 of *E. coli* BL21-pET23b(+)-NICH1 still maintained excellent hydrolysis ability at 7th day, and the COS yield reached 41%. Compared with crude CHI-1, *E. coli* BL21-pET23b(+)-NICH1 can be buffered into a high concentration of bacterial suspension without being separated and purified to concentrate. At room temperature, the engineered *E. coli* BL21-pET23b(+)-NICH1 suspension still had high stability after 40 days. The fermentation of *B. subtilis* was performed for 3 days, followed by co-fermented with *Acetobacter* sp. for another 5 days. The total process time for chitin extraction was 8 days, and the final DP%, DM%, and chitin yield were 94, 92 and 18%, respectively. The average molecular weight of crude chitin was 54.6 kDa, and the DD reached 13%. After the deacetylation, chitosan with an average molecular weight of 11.1 kDa and a DD of 95% was prepared. In a word, a new process for extracting COS from shrimp chaff is established, which has prospective biotechnology application in COS preparation.

Abbreviations

DP deproteinization

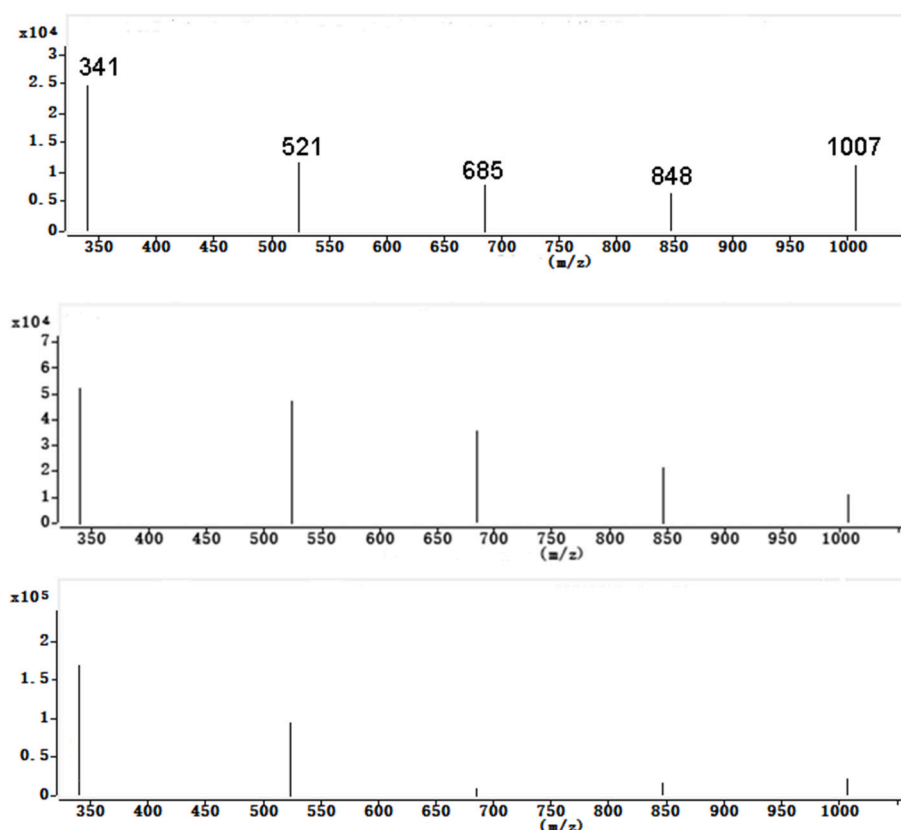


Fig. 7. LC-MS analysis of products for hydrolytic products of prepared chitosan by crude CHI-1 for 3 days (A), and *E. coli* BL21-pET23b(+)-NICH1 (B, C) for 3 days and 7 days, respectively. The m/z values were accounted for the nominal masses of the ions $[M + H]^+$.

DM	demineralization
PBS	phosphate buffered saline
LB	Luria-Bertani
TTA	titratable acid
DNS	dinitrosalicylic acid
COS	chitooligosaccharides
DD	deacetylation degree
LC-MS	liquid chromatography and mass spectrometry
SEM	scanning electron microscope
FTIR	Fourier transform infrared spectroscopy
GH	glycoside hydrolase
GlcN2	chitobiose
GlcN3	chitotriose
GlcN4	chitotetraose
GlcN5	chitopentaose
GlcN6	chitohexaose
InaK-N	N-terminal region of the ice nucleation protein
CFU	colony-forming unit
DMAC	<i>N, N</i> -dimethylacetamide
MHS	Mark-Houwink-Sakurada

CRediT authorship contribution statement

Yanli Chen: Conceptualization, Methodology, Writing – original draft. **Zhenmin Ling:** Conceptualization, Writing – original draft. **Tur-sunay Mamtimin:** Supervision. **Aman Khan:** Supervision. **Liang Peng:** Software. **Jinfeng Yang:** Software. **Gohar Ali:** Visualization. **Tuoyu Zhou:** Investigation. **Qing Zhang:** Methodology. **Jing Zhang:** Methodology. **Xiangkai Li:** Conceptualization, Writing – review & editing, Validation, Project administration.

Declaration of competing interest

Authors declare that there are no conflicts of interest regarding the publication of this paper.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (No. 31870082 and 32070117), Lanzhou Science and Technology Plan Project [grant numbers: 2019-4-40], National Natural Science Foundation of China [No: 31600079], Natural Science Foundation of Gansu Province [No: 20JR5RA234] and Fundamental Research Funds for the Central Universities [No: lzujbky-2020-kb05]. We thank the School of Life and Sciences, Lanzhou University, for providing instruments and equipment. We also wish to thank the Electron Microscopy Centre of Lanzhou University for microscopy and microanalysis of our specimens.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbpol.2021.118894>.

References

- Affes, S., Aranaz, I., Hamdi, M., Acosta, N., Ghorbel-Bellaaj, O., Heras, A., & Maalej, H. (2019). Preparation of a crude chitosanase from blue crab viscera as well as its application in the production of biologically active chito-oligosaccharides from shrimp shells chitosan. *International Journal of Biological Macromolecules*, 139, 558–569. <https://doi.org/10.1016/j.ijbiomac.2019.07.116>
- Ali, G., Ling, Z., Saif, I., Usman, M., Jalalah, M., Harraz, F. A., & Li, X. (2021). Biomethanation and microbial community response during agricultural biomass and shrimp chaff digestion. *Environmental Pollution*, 278, Article 116801. <https://doi.org/10.1016/j.envpol.2021.116801>

- Azevedo, M. I. G., Oliveira, S. T., Silva, C. F. B., Carneiro, R. F., Nagano, C. S., Gadelha, A. C. S., & Grangeiro, T. B. (2020). Secretory production in *Escherichia coli* of a GH46 chitosanase from *Chromobacterium violaceum*, suitable to generate antifungal chitooligosaccharides. *International Journal of Biological Macromolecules*, 165(Pt A), 1482–1495. <https://doi.org/10.1016/j.ijbiomac.2020.09.221>
- Chatalein, P. G., Pintado, M. E., & Vasconcelos, M. W. (2014). Evaluation of chitooligosaccharide application on mineral accumulation and plant growth in *Phaseolus vulgaris*. *Plant Science*, 215, 134–140. <https://doi.org/10.1016/j.plantsci.2013.11.009>
- Choi, Y. J., Kim, E. J., Piao, Z., Yun, Y. C., & Shin, Y. C. (2004). Purification and characterization of chitosanase from *Bacillus* sp. Strain KCTC 0377BP and its application for the production of chitosan oligosaccharides. *Applied and Environmental Microbiology*, 70(8), 4522–4531. <https://doi.org/10.1128/AEM.70.8.4522-4531.2004>
- De Araujo Padilha, C. E., Fortunato Dantas, P. V., de Sousa, F. C. J., de Santana Souza, D. F., de Oliveira, J. A., de Macedo, G. R., & Dos Santos, E. S. (2016). Mathematical modeling of the whole expanded bed adsorption process to recover and purify chitosanases from the unclarified fermentation broth of *paenibacillus ehimensis*. *Journal of Chromatography. B, Analytical Technologies in the Biomedical and Life Sciences*, 1039, 44–50. <https://doi.org/10.1016/j.jchromb.2016.10.027>
- Dimassi, S., Tabary, N., Chai, F., Blanchemain, N., & Martel, B. (2018). Sulfonated and sulfated chitosan derivatives for biomedical applications: A review. *Carbohydrate Polymers*, 202, 382–396. <https://doi.org/10.1016/j.carbpol.2018.09.011>
- Ding, M., Zhang, T., Sun, C., Zhang, H., & Zhang, Y. (2020). A chitosanase mutant from *Streptomyces* sp. N174 prefers to produce functional chitopentasaaccharide. *International Journal of Biological Macromolecules*, 151, 1091–1098. <https://doi.org/10.1016/j.ijbiomac.2019.10.151>
- Dun, Y., Li, Y., Xu, J., Hu, Y., Zhang, C., Liang, Y., & Zhao, S. (2019). Simultaneous fermentation and hydrolysis to extract chitin from crayfish shell waste. *International Journal of Biological Macromolecules*, 123, 420–426. <https://doi.org/10.1016/j.ijbiomac.2018.11.088>
- Gabriel Paulraj, M., Ignacimuthu, S., Gandhi, M. R., Shajahan, A., Ganesan, P., Packiam, S. M., & Al-Dhabi, N. A. (2017). Comparative studies of tripolyphosphate and glutaraldehyde cross-linked chitosan-bacterial pesticide nanoparticles and their agricultural applications. *International Journal of Biological Macromolecules*, 104(Pt B), 1813–1819. <https://doi.org/10.1016/j.ijbiomac.2017.06.043>
- Germec, M., & Turhan, I. (2019). Evaluation of carbon sources for the production of inulinase by *Aspergillus Niger* A42 and its characterization. *Bioprocess and Biosystems Engineering*, 42(12), 1993–2005. <https://doi.org/10.1007/s00449-019-02192-9>
- Germec, M., & Turhan, I. (2020). Thermotability of *Aspergillus niger* inulinase from sugar beet molasses in the submerged fermentation and determination of its kinetic and thermodynamic parameters. *Biomass Conversion and Biorefinery*. <https://doi.org/10.1007/s13399-020-00809-8>
- Ghorbel-Bellaaj, O., Younes, I., Maalej, H., Hajji, S., & Nasri, M. (2012). Chitin extraction from shrimp shell waste using bacillus bacteria. *International Journal of Biological Macromolecules*, 51(5), 1196–1201. <https://doi.org/10.1016/j.ijbiomac.2012.08.034>
- Guan, F., Han, Y., Yan, K., Zhang, Y., Zhang, Z., Wu, N., & Tian, J. (2020). Highly efficient production of chitooligosaccharides by enzymes mined directly from the marine metagenome. *Carbohydrate Polymers*, 234, Article 115909. <https://doi.org/10.1016/j.carbpol.2020.115909>
- Guo, N., Sun, J., Wang, W., Gao, L., Liu, J., Liu, Z., & Mao, X. (2019). Cloning, expression and characterization of a novel chitosanase from *Streptomyces albolongus* ATCC 27414. *Food Chemistry*, 286, 696–702. <https://doi.org/10.1016/j.foodchem.2019.02.056>
- Ismail, S. A., El-Sayed, H. S., & Fayed, B. (2020). Production of prebiotic chitooligosaccharide and its nano/microencapsulation for the production of functional yoghurt. *Carbohydrate Polymers*, 234. <https://doi.org/10.1016/j.carbpol.2020.115909>
- Jung, W. J., & Park, R. D. (2014). Bioproduction of chitooligosaccharides: Present and perspectives. *Marine Drugs*, 12(11), 5328–5356. <https://doi.org/10.3390/md12115328>
- Kavaz, D., Idris, M., & Onyebuchi, C. (2019). Physicochemical characterization, antioxidative, anticancer cells proliferation and food pathogens antibacterial activity of chitosan nanoparticles loaded with *Cyperus articulatus* rhizome essential oils. *International Journal of Biological Macromolecules*, 123, 837–845. <https://doi.org/10.1016/j.ijbiomac.2018.11.177>
- Kuroiwa, T., Izuta, H., Nabetani, H., Nakajima, M., Sato, S., Mukataka, S., & Ichikawa, S. (2009). Selective and stable production of physiologically active chitosan oligosaccharides using an enzymatic membrane bioreactor. *Process Biochemistry*, 44(3), 283–287. <https://doi.org/10.1016/j.procbio.2008.10.020>
- Li, N., He, Y., Lian, J., Liu, Q. Y., Zhang, Y. X., & Zhang, X. (2020). Hg(2+) significantly enhancing the peroxidase-like activity of H2TCP/ZnS/CoS nanoparticles by inducing the formation of surface-cation defects and application for the sensitive and selective detection of Hg(2+) in the environment. *Inorganic Chemistry*, 59(24), 18384–18395. <https://doi.org/10.1021/acs.inorgchem.0c03007>
- Li, Q., Wang, T., Ye, Y., Guan, S., Cai, B., Zhang, S., & Rong, S. (2021). A temperature-induced chitosanase bacterial cell-surface display system for the efficient production of chitooligosaccharides. *Biotechnology Letters*, 43(8), 1625–1635. <https://doi.org/10.1007/s10529-021-03139-5>
- Liu, M., Lu, X., Khan, A., Ling, Z., Wang, P., Tang, Y., & Li, X. (2019). Reducing methylmercury accumulation in fish using *Escherichia coli* with surface-displayed methylmercury-binding peptides. *Journal of Hazardous Materials*, 367, 35–42. <https://doi.org/10.1016/j.jhazmat.2018.12.058>
- Lombard, V., Golaconda Ramulu, H., Drula, E., Coutinho, P. M., & Henrissat, B. (2014). The carbohydrate-active enzymes database (CAZy) in 2013. *Nucleic Acids Research*, 42(Database issue), D490–D495. <https://doi.org/10.1093/nar/gkt1178>
- Mathew, G. M., Mathew, D. C., Sukumaran, R. K., Sindhu, R., Huang, C. C., Binod, P., & Pandey, A. (2020). Sustainable and eco-friendly strategies for shrimp shell valorization. *Environmental Pollution*, 267, Article 115656. <https://doi.org/10.1016/j.envpol.2020.115656>
- Monaghan, R. L., Eveleigh, D. E., Tewari, R. P., & Reese, E. T. (1973). Chitosanase, a novel enzyme. *Nature: New Biology*, 245(142), 78–80. <https://doi.org/10.1038/newbio245078a0>
- Muanprasat, C., & Chatsudthipong, V. (2017). Chitosan oligosaccharide: Biological activities and potential therapeutic applications. *Pharmacology & Therapeutics*, 170, 80–97. <https://doi.org/10.1016/j.pharmthera.2016.10.013>
- Naqvi, S., Cord-Landwehr, S., Singh, R., Bernard, F., Kolkenbrock, S., El Gueddari, N. E., & Moerschbacher, B. M. (2016). A recombinant fungal chitin deacetylase produces fully defined chitosan oligomers with novel patterns of acetylation. *Applied and Environmental Microbiology*, 82(22), 6645–6655. <https://doi.org/10.1128/AEM.01961-16>
- Nidheesh, T., Pal, G. K., & Suresh, P. V. (2015). Chitooligomers preparation by chitosanase produced under solid state fermentation using shrimp by-products as substrate. *Carbohydrate Polymers*, 121, 1–9. <https://doi.org/10.1016/j.carbpol.2014.12.017>
- Nwe, N., Furuie, T., & Tamura, H. (2014). Isolation and characterization of chitin and chitosan from marine origin. *Advances in Food and Nutrition Research*, 72, 1–15. <https://doi.org/10.1016/B978-0-12-800269-8.00001-4>
- Park, P. J., Je, J. Y., & Kim, S. K. (2004). Free radical scavenging activities of differently deacetylated chitosans using an ESR spectrometer. *Carbohydrate Polymers*, 55(1), 17–22. <https://doi.org/10.1016/j.carbpol.2003.05.002>
- Rywinska, A., Marcinkiewicz, M., Cibis, E., & Rymowicz, W. (2015). Optimization of medium composition for erythritol production from glycerol by *Yarrowia lipolytica* using response surface methodology. *Preparative Biochemistry & Biotechnology*, 45(6), 515–529. <https://doi.org/10.1080/10826068.2014.940966>
- Sedaghat, F., Yousefzadi, M., Toiserkani, H., & Najafipour, S. (2017). Bioconversion of shrimp waste *Penaeus merguensis* using lactic acid fermentation: An alternative procedure for chemical extraction of chitin and chitosan. *International Journal of Biological Macromolecules*, 104(Pt A), 883–888. <https://doi.org/10.1016/j.ijbiomac.2017.06.099>
- Sinha, S., Chand, S., & Tripathi, P. (2016). Recent progress in chitosanase production of monomer-free chitooligosaccharides: Bioprocess strategies and future applications. *Applied Biochemistry and Biotechnology*, 180(5), 883–899. <https://doi.org/10.1007/s12010-016-2140-6>
- Teli, M. D., & Sheikh, J. (2012). Extraction of chitosan from shrimp shells waste and application in antibacterial finishing of bamboo rayon. *International Journal of Biological Macromolecules*, 50(5), 1195–1200. <https://doi.org/10.1016/j.ijbiomac.2012.04.003>
- Wang, W., Bo, S. Q., Li, S. Q., & Qin, W. (1991). Determination of the mark-houwink equation for chitosans with different degrees of deacetylation. *International Journal of Biological Macromolecules*, 13(5), 281–285. [https://doi.org/10.1016/0141-8130\(91\)90027-r](https://doi.org/10.1016/0141-8130(91)90027-r)
- Wieckiewicz, M., Boening, K. W., Grychowska, N., & Paradowska-Stolarz, A. (2017). Clinical application of chitosan in dental specialties. *Mini Reviews in Medicinal Chemistry*, 17(5), 401–409. <https://doi.org/10.2174/1389557516666160418123054>
- Xiong, C., Wu, H., Wei, P., Pan, M., Tuo, Y., Kusakabe, I., & Du, Y. (2009). Potent angiogenic inhibition effects of deacetylated chitohexaose separated from chitooligosaccharides and its mechanism of action in vitro. *Carbohydrate Research*, 344(15), 1975–1983. <https://doi.org/10.1016/j.carres.2009.06.036>
- Xu, W., Mohan, A., Pitts, N. L., Udenigwe, C., & Mason, B. (2020). Bile acid-binding capacity of lobster shell-derived chitin, chitosan and chitooligosaccharides. *Food Bioscience*, 33. <https://doi.org/10.1016/j.fbio.2019.100476>
- Yang, J. Y., Zhang, H., Chen, J., Chen, X. Q., Wang, S., Li, Y. L., & Kong, Y. (2016). The application of mandarin acceptable noise level and COS in hearing aid fitting for presbycusis. *Lin Chung Er Bi Yan Hou Tou Jing Wai Ke Za Zhi*, 30(23), 1850–1853. <https://doi.org/10.13201/j.issn.1001-1781.2016.23.004>, 1858.
- Younes, I., & Rinaudo, M. (2015). Chitin and chitosan preparation from marine sources. Structure, properties and applications. *Marine Drugs*, 13(3), 1133–1174. <https://doi.org/10.3390/md13031133>
- Yuan, Y., Xu, X., Gong, J., Mu, R., Li, Y., Wu, C., & Pang, J. (2019). Fabrication of chitosan-coated konjac glucomannan/sodium alginate/graphene oxide microspheres with enhanced colon-targeted delivery. *International Journal of Biological Macromolecules*, 131, 209–217. <https://doi.org/10.1016/j.ijbiomac.2019.03.061>
- Zhang, J., Cao, H., Li, S., Zhao, Y., Wang, W., Xu, Q., & Yin, H. (2015). Characterization of a new family 75 chitosanase from *Aspergillus* sp. W-2. *International Journal of Biological Macromolecules*, 81, 362–369. <https://doi.org/10.1016/j.ijbiomac.2015.08.026>
- Zhang, Q., Wang, L., Liu, S., & Li, Y. (2021). Establishment of successive co-fermentation by *Bacillus subtilis* and *Acetobacter pasteurianus* for extracting chitin from shrimp shells. *Carbohydrate Polymers*, 258, Article 117720. <https://doi.org/10.1016/j.carbpol.2021.117720>
- Zhang, X. Q., Li, K. C., Liu, S., Zou, P., Xing, R. E., Yu, H. H., & Li, P. C. (2018). Relationship between the degree of polymerization of chitooligomers and their activity affecting the growth of wheat seedlings under salt stress (vol 65, pg 501, 2017). *Journal of Agricultural and Food Chemistry*, 66(28). <https://doi.org/10.1021/acs.jafc.8b03268>, 7551–7551.

- Zheng, L. Y., & Xiao, Y. L. (2004). Penicillium sp ZD-Z1 chitosanase immobilized on DEAE cellulose by cross-linking reaction. *Korean Journal of Chemical Engineering*, 21(1), 201–205.
- Zhou, J. L., Liu, X. B., Yuan, F., Deng, B., & Yu, X. B. (2020). Biocatalysis of heterogenously-expressed chitosanase for the preparation of desirable chitosan oligosaccharides applied against phytopathogenic fungi. *ACS Sustainable Chemistry & Engineering*, 8(12), 4781–4791. <https://doi.org/10.1021/acssuschemeng.9b07288>
- Zhu, X. Y., Zhao, Y., Zhang, H. D., Wang, W. X., Cong, H. H., & Yin, H. (2019). Characterization of the specific mode of action of a chitin deacetylase and separation of the partially acetylated chitosan oligosaccharides. *Marine Drugs*, 17(2). <https://doi.org/10.3390/md17020074>